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Domain- Artificial Intelligence and Machine Learning

Duration- 4 weeks

Batch start date- 25 may

Github-

<https://github.com/RajGhule-10/sumerix-global-ai-ml-internship-raj-ghule/tree/main>

TASK 1 REPORT

HOUSE PRICE PREDICTION

→ using california housing dataset

TITLE

House Price Prediction using Linear Regression

INTRODUCTION

Machine Learning is widely used to analyze data and make predictions. In this project, a Linear Regression model was developed using the California Housing dataset to predict house prices based on features such as income, house age, rooms, and population. The project demonstrates the complete machine learning workflow including data loading, exploration, visualization, model training, prediction, and evaluation.

OBJECTIVE

The objective of this project is to build and evaluate a Linear Regression model using the California Housing dataset. The model predicts house prices based on different housing features.

TECHNOLOGIES USED

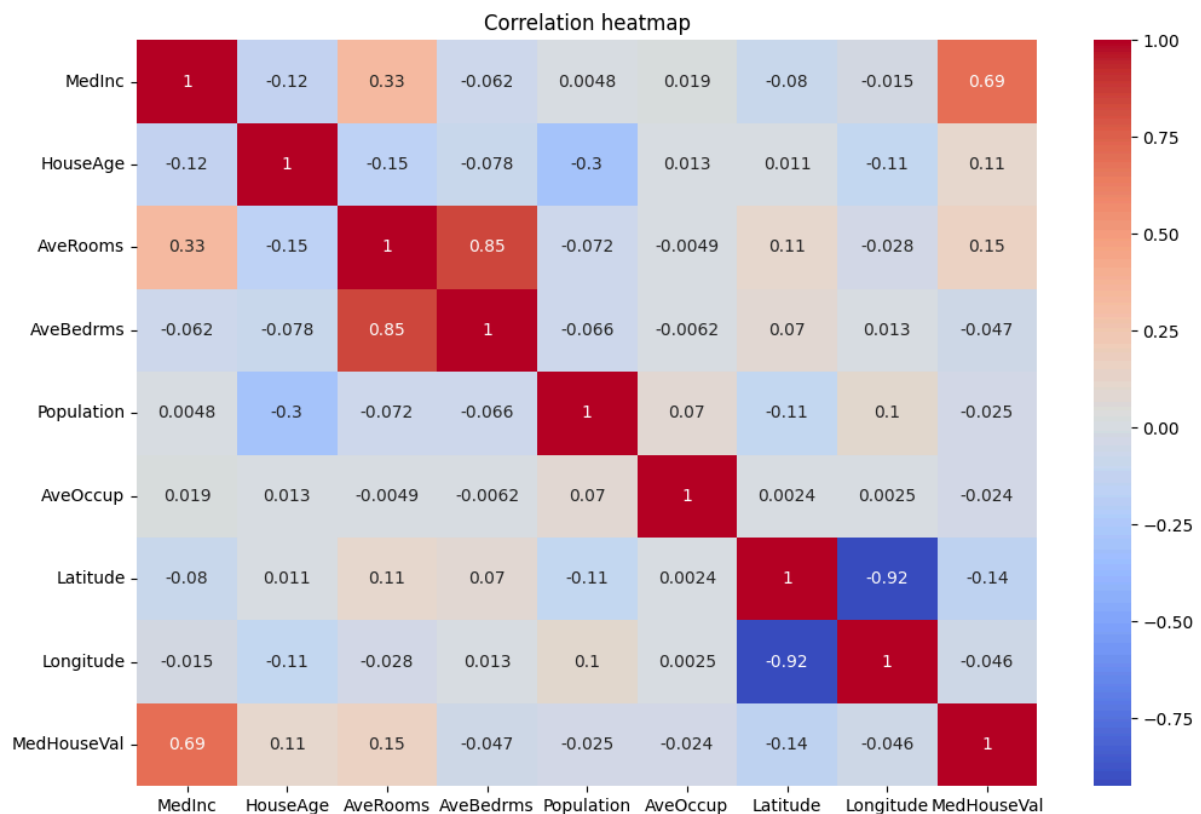
The following technologies and libraries were used in this project:

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- Jupyter Notebook

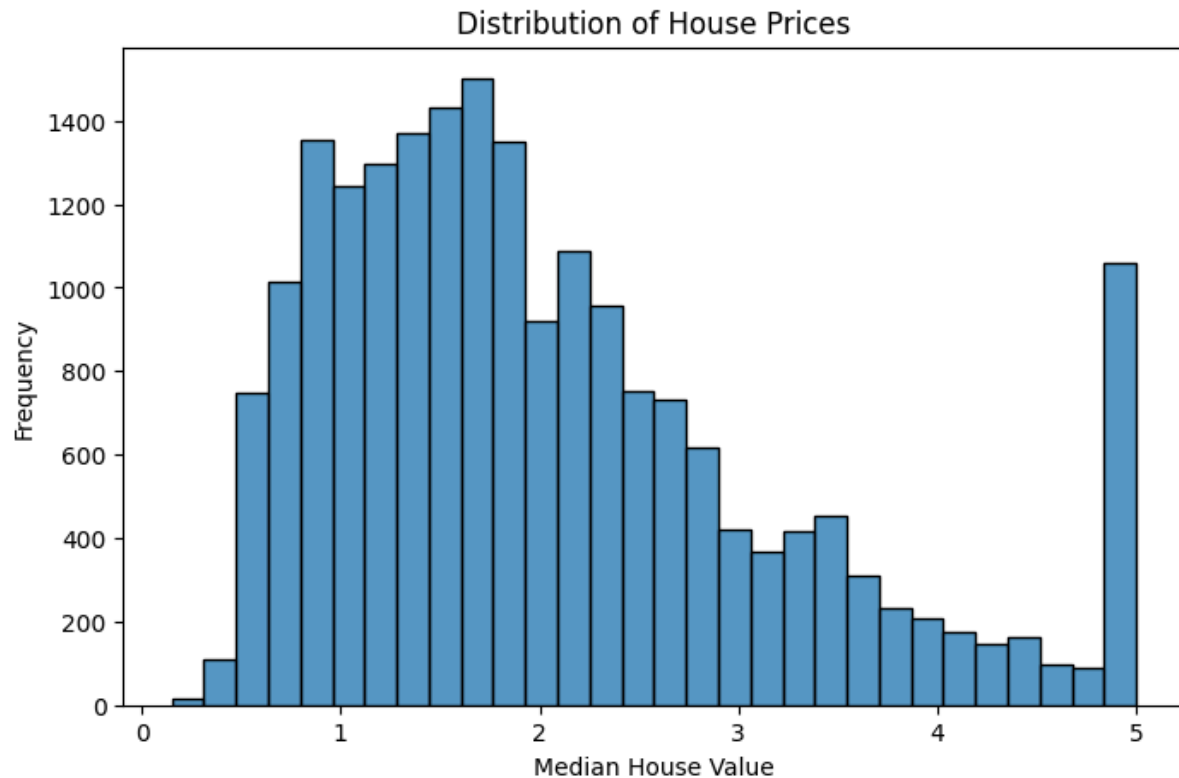
DATASET INFORMATION

The California Housing dataset was imported from Scikit-learn. It contains information such as median income, house age, average rooms, population, and house prices.

DATA EXPLORATION AND VISUALIZATION



→ The correlation heatmap shows relationships between different housing features. Median income has a strong positive correlation with house prices, indicating that areas with higher income generally have higher house values.



The histogram shows the distribution of house prices in the dataset. Most house prices are concentrated within a moderate price range.

MODEL TRAINING AND EVALUATION

Model used- Linear regression

The dataset was divided into training and testing data using an 80-20 split ratio.

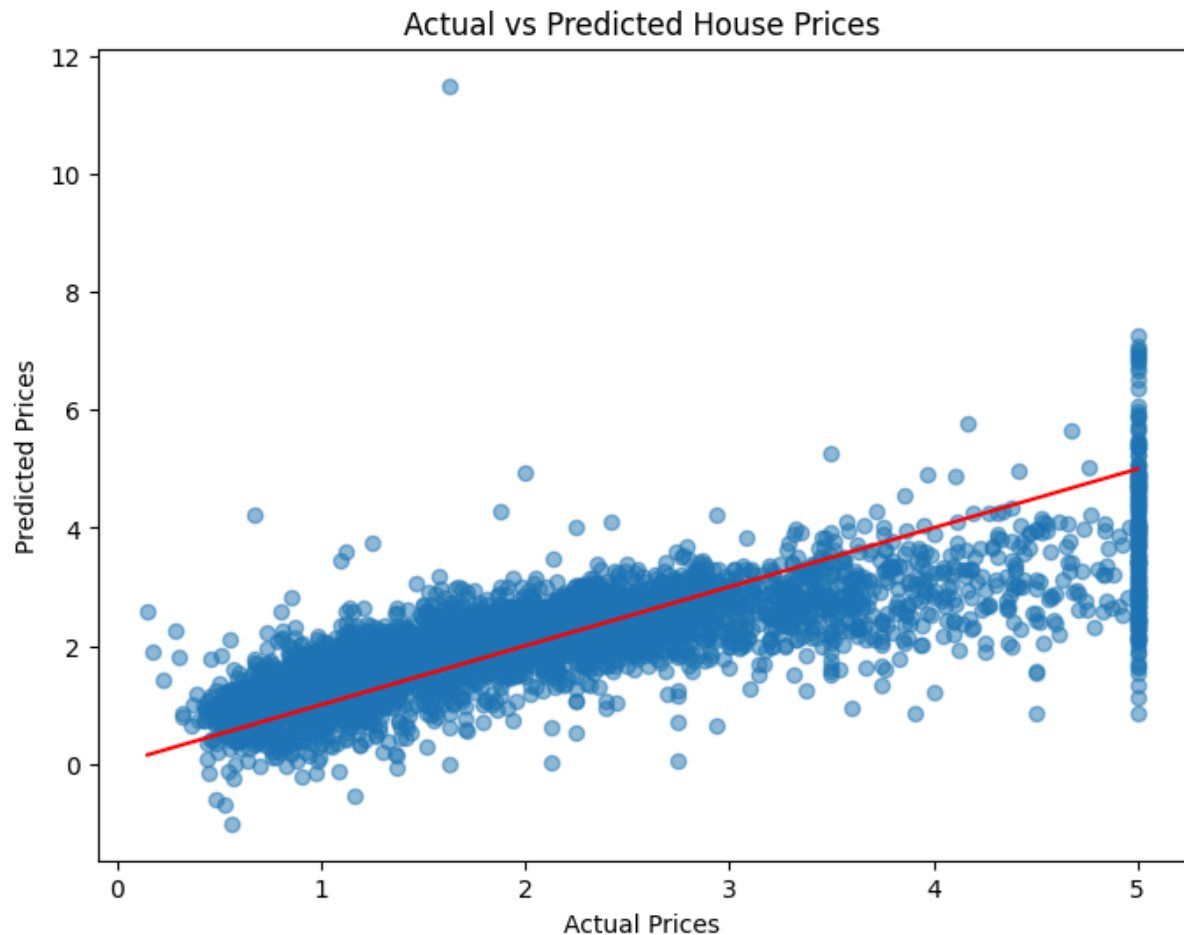
METRIC VALUES-

Model Evaluation Metrics

Mean Absolute Error (MAE): 0.533

Root Mean Squared Error (RMSE): 0.746

R2 Score: 0.576



The model was able to predict house prices reasonably well. The Actual vs Predicted graph shows that many predictions are close to the actual values.

CONCLUSION-

In this project, a Linear Regression model was successfully built and evaluated using the California Housing dataset. The project helped in understanding the complete machine learning workflow including data preprocessing, visualization, model training, prediction, and evaluation.

FUTURE IMPROVEMENTS-

The model performance can be improved further using advanced machine learning algorithms such as Random Forest, XGBoost, or Gradient Boosting. Feature engineering and hyperparameter tuning can also improve prediction accuracy.